

Examiner reports

Q1.

In part (a) most candidates found the correct values of a and b , but correct values for r^2 were not so common. Some sloppiness was again evident with candidates failing to write squared outside the brackets or omitting the plus sign between the terms on the left hand side. It was common to see things such as $25 = \sqrt{25} = 5^2$ and this could be penalised in the future.

In part (b) the coordinates of the centre, C and the radius r , although not always correct, usually gained full credit when following through from part (a).

In part (c)(i) most candidates attempted to verify that the point D was on the circle, although some, who had obviously worked a previous examination question, were keen to show that the distance from C to D was less than the radius and that D lay inside the circle. This verification was marked fairly strictly and the argument had to be correct including a final concluding statement. Those who simply wrote $4^2 + 3^2 = 25$, for example, did not earn the mark.

In part (c)(ii) many candidates found the gradient of CD and then assumed they had to find the negative reciprocal of this since the question asked for the normal at D . Reference to a sketch might have prevented this incorrect assumption.

In part (d)(i) most candidates made errors by not using brackets; the expression kx^2 was seen almost as often as the correct form k^2x^2 after substituting $y = kx$ into their circle equation.

In part (d)(ii), although there were some correct solutions seen, the discriminant often contained algebraic slips and the condition for equal roots was rarely stated. Often it was several lines into the working before an “= 0” appeared and many times this was omitted entirely. The value $k = 0$ was often ignored in otherwise correct solutions, but it was more common to see a three term quadratic because of previous algebraic errors.

In part (d)(iii) several candidates realised that the line would be a tangent for each of the two values of k , but many completely missed the point and talked about transformations, often giving vectors in their answer.

Q2.

In part (a)(i) many candidates were unable to make y the subject of the equation $3x + 5y = 11$ and, as a result, many incorrect answers for the gradient were seen. Those who tried to use two points on the line to find the gradient were rarely successful.

In part (a)(ii) most candidates realised that the product of the gradients of perpendicular lines should be -1 and credit was given for using this result together with their answer from part(a)(i).

Although many correct answers for the coordinates of C were seen in part (b)(i), the simultaneous equations defeated a large number of candidates. No credit was given for mistakenly using their equation from part (a)(ii) instead of the correct equation for AB .

Q3.

In part (a)(i) most candidates realised what the correct coordinates of the centre were, although some wrote these as $(-5, 12)$ instead of $(5, -12)$.

Some gave the radius as 169 and others evaluated $\sqrt{169}$ incorrectly in part (a)(ii). The majority of candidates obtained the correct value of the radius.

In part (b)(i) most were able to verify that the circle passed through the origin, although some neglected to make a statement as a conclusion to their calculation and so failed to earn this mark. A surprisingly large number made no attempt at this part.

Most sketches were correct in part (b)(ii), though some were very untidy with some making several attempts at the circle so the diagram resembled the chaotic orbit of a planet. In spite of being asked to verify that the circle passed through the origin many sketches did not do so. Credit was given for freehand circles with the centre in the correct quadrant and which passed through the origin, although it was good to see some circles drawn using compasses. Many used algebraic methods, putting $x = 0$, but often their poor algebra prevented them from finding the value of p . Those using the symmetry, doubling the y -coordinate, were usually more successful, although an answer of -25 (from $-12-13$) was common.

In part (c)(i) the majority of candidates tried to find the gradient of AC but careless arithmetic meant that far fewer actually succeeded in finding its correct simplified value.

In part (c)(ii), in order to find the tangent, it was necessary to use the negative reciprocal of the answer from part (c)(i) in order to find the gradient. Although some did, many chose to use the same gradient obtained in the previous part of the question and scored no marks at all.

Q4.

In part (a) almost all candidates obtained $dx/dt = -1/t^2$. However, most candidates could not handle the fraction and indices correctly when finding dy/dt . The most common mistake was to state $dy/dt = 1 - 2/t^2$. Sometimes this “became” $1 - 1/2t^2$ but, by then, the mark had been lost. The t^2 sometimes migrated on to the numerator. Almost all candidates knew they needed to divide dy/dt by dx/dt to get dy/dx . Full marks were awarded for an unsimplified form of dy/dx .

A few candidates confused differentiation and integration giving the “derivatives” in terms of $\ln t$. Some candidates proceeded by first attempting to find the cartesian equation, which was valid if the result was given in terms of t . Surprisingly many of these candidates then didn’t make much progress in part (c). In part (b) the majority of candidates found the equation of the normal, and not the tangent, and the relationship between the gradients was well known.

Very few found an incorrect y value at $t = 1$. Most errors were due to an incorrect dy/dx from part (a), but some credit was given for method and 3 marks were still available. These were earned by most. Some candidates had an expression for dy/dt in which the t ’s cancelled, yet it didn’t apparently occur to them that an error had been made. Most candidates began by writing the cartesian equation in an unsimplified form in part (c). Many were defeated by the fraction in the denominator and so could not cope with

$1/(2(1/x))$. Errors were often made when attempting to multiply through by $2x$. Some candidates chose to substitute for t in the given equation, but not all could deal with the algebra involved; $k = 1$ or $k = -2$ were common wrong conclusions. Pleasingly most candidates who did attempt this question made an algebraic approach and very few just substituted $x = 1$ and $y = 3/2$ to find the value of k , which gained no credit.

Q5.

In part (a), apart from a few sign errors, it was pleasing to see that most candidates were able to find the correct mid-point. However, those who insisted on subtracting the coordinates before dividing by 2 would do well to learn the formula in the first bullet point above. Quite a few candidates found the mid-point of AB instead of BC , and this was generously treated as a misread.

In part (b)(i), many ignored the request to simplify the gradient, but most were successful

in writing the gradient of AB as $-\frac{1}{3}$.

In part (b)(ii), almost all candidates managed to write down a correct equation for the line AB , but careless arithmetic prevented many from obtaining the required form of $x + 3y = 7$. Some were content to give a final answer that was not in the required form, thus losing a mark.

In part (b)(iii), some candidates immediately used $m_1 \times m_2 = -1$ to find the gradient of the parallel line and scored no marks. Many who used the formula $y = mx + c$ for the equation of the straight line through C parallel to AB made arithmetic slips and did not obtain a correct final equation.

In part (c), the most common approach, and the one expected, was to use gradients in order to prove that angle ABC was a right angle. Some simply assumed the result, stating

that since the gradient of AB was $-\frac{1}{3}$ then BC had gradient 3. It was necessary to show, by considering the differences of the coordinates that BC had gradient 3. Far too many simply found the two gradients and wrote “therefore the lines BC and AB are perpendicular”. Since this was a proof, it was expected that the product of the two gradients would be shown to equal -1 before a statement was made about angle ABC being a right angle. Some were successful in proving the result using Pythagoras’ Theorem, but many attempts were incomplete with several candidates writing

$\sqrt{40} + \sqrt{40} = \sqrt{80}$ or other inaccurate statements. Others used the cosine rule, and one or two used the scalar product of two vectors in order to prove the result. A surprising number confused “isosceles” with “right-angled” and, having found two equal sides, stated that the result was proved.

Q6.

In part (a)(i), a few candidates ignored the request to use the factor theorem and scored no marks for using long division. It was necessary to make a statement that “ $x + 1$ is a factor”, after showing that $f(-1) = 0$, in order to score full marks.

Part (a)(ii) was not answered as well as similar questions in previous years. Perhaps the

sketch lured some into trying to write down three factors without any further working, rather than using the intermediate step of showing that $p(x) = (x + 1)(x^2 - x - 6)$ before writing $p(x)$ as a product of three factors. Many who tried long division were flummoxed by there being no x^2 term.

In part (b)(i), those who had the correct linear factors in part (a)(ii) usually wrote down correctly that A had coordinates $(-2, 0)$, although some carelessly wrote the point as $(0, -2)$.

Many candidates simply found an indefinite integral in part (b)(ii) and then a definite integral in part (b)(iii). The two parts were generously treated holistically when candidates did this. The fractions once again caused problems to most candidates who are so used to having a calculator to do this work for them. It was very rare to see the correct answer of -32 for the definite integral.

In part (b)(iii), many lost out on an easy mark because they rolled their two sections into one: those who wrote “integral = $-32 = 32$ ” gained full credit for part (b)(ii) but did not score the mark in part (b)(iii). It was necessary to give a positive value for the area of the region and to make this explicit. In anticipation of a lot of wrong answers in part (b)(ii), a follow through mark was awarded in part (b)(iii): for example, if a candidate’s answer in part (b)(ii) was -20 and they concluded that the area was 20 in part (b)(iii), they scored the mark.

In part (b)(iv), most candidates differentiated correctly, but quite a few thought that $3(-1)^2 - 7$ was equal to -10 and thus obtained the wrong gradient of the curve.

In part (b)(v), a large number of candidates found the correct equation of the normal but some still confused tangents and normals and consequently thought that the gradient of the normal was equal to -4 . It was quite common for weaker candidates to either negate their gradient or take the reciprocal but to fail to do both.

Q7.

Generally, most candidates scored relatively high marks for this long structured calculus question. Parts (a)(i), (ii) and (iii) were answered very well, although some used the gradient of the normal in part (a)(iii) as -2 and so in part (a)(iv) obtained a positive value for the y -coordinate of Q . A glance at the diagram should have indicated that such a value was incorrect.

Part (a)(v), which required candidates to find the x -coordinate of the maximum point, M , was generally not answered with any confidence. To score any marks for this part, candidates were required to go beyond just equating their answer for part (a)(i) to zero. Many candidates again incorrectly believed that for a maximum point the second derivative was zero. Candidates generally found the correct answer to the indefinite integral in part (b)(i), but a significant minority forgot to add the area of the triangle to their ‘area under the curve’ in part (b)(ii) of the question.

Q8.

Part (a) was usually answered well and several correct equations were seen. However

errors such as $\frac{-15}{-3} = -5$ sometimes occurred, and weaker candidates confused coordinates and wrote equations such as $y - 15 = m(x + 2)$. Some attempted to

differentiate to find the gradient instead of using the coordinates of the points.

Although the integration in part (b)(i) was largely successful, the evaluation of $F(1) - F(-2)$ defeated most. Arithmetic errors, usually caused by minus signs and fractions, abounded.

Some assumed that they had answered part (b)(ii) by substituting the limits after integration. Of those who realised the need to subtract the area of a triangle, many assumed the triangle base to be 4 rather than 3. Another error seen was the integration of the curve from -2 to $+2$ followed by subtracting their answer to part (b)(i). Some successfully found the area under the line by integration, but that was not the expected method.

Q9.

In part (a), while many found the centre and hence the values of a and b correctly, far fewer were able to find the radius. A very common incorrect answer was $\sqrt{233}$ instead of 13. A few candidates substituted the coordinates of the centre into the given equation so wrote $(8 - a)^2 + (13 - b)^2 = \dots$ and earned no marks. Too many candidates were drawn into fruitless calculations when all they needed to do was refer to the diagram.

In part (b)(i), the gradient of PC was usually correct.

The most common error in part (b)(ii) was a misunderstanding of what was required. Many assumed that the tangent was the straight line passing through P and C and used their answer to part (b), instead of its negative reciprocal, as the gradient. Not all of those who found the correct equation converted it to a form with integer coefficients.

In part (b)(iii), a surprising number of candidates did not realise that the distance from C to P was the radius, and recalculated it. It was a pity that they did not return to part (a) and correct their circle equation.

Those who drew a sketch usually used Pythagoras's Theorem correctly, but many failed to recognise that the distance required was the length of the perpendicular from the centre of the circle to the midpoint of the chord. A few obtained the correct answer by considering a horizontal chord and finding the point of intersection with the circle.

Q10.

Although many candidates were able to correctly write $\sqrt{x^3}$ in the form $x^{\frac{3}{2}}$ it was not uncommon to see the incorrect answers $k = 3.5$ or $k = \frac{1}{3}$. Although follow through marks were available, a significant minority of candidates subsequently differentiated $x^{\frac{1}{3}}$ incorrectly as $\frac{1}{3}x^{\frac{2}{3}}$. In addition to arithmetical errors the most common error in finding the equation of the tangent was to use the gradient of the tangent as $-\frac{1}{5}$; in effect, the equation of the normal was found. Very few candidates failed to write their equation in the required form.

Q11.

The vast majority of candidates showed they understood the requirements of this question although often algebraic errors were made.

Many candidates could not find the derivative of $\frac{1}{2t}$ correctly in part (a), the 2 moving between the numerator and denominator position to various powers; the derivative of $\frac{1}{t}$ itself was usually correct. However, most candidates used the chain rule correctly to find a gradient. Some candidates actually found a cartesian equation before differentiating, with some completing successfully.

In part (b) most candidates went on to use their gradient to find that of the normal, and bar a few slips in calculating the x and y coordinates, were successful in finding a correct equation, using their gradient.

In part (c) most candidates found t from $x = 4t + 3$ and substituted in $y = \frac{1}{2t} - 1$, often without any simplification, which was quite acceptable. Those who did attempt some simplification often made algebraic errors, but these were condoned following a correct substitution. Those who attempted an alternative method of eliminating t were more prone

to error. For some candidates, poor presentation led to $\frac{1}{2t}$ becoming $\frac{1}{2}t$.

Q12.

Part (a)(i) It was disappointing to see many candidates unable to rearrange $3x + 5y = 8$ to make y the subject in order to find the gradient. Some were successful in finding a second point on the line such as (1, 1) and then using the coordinates of A to find the gradient of AB .

Part (a)(ii) Most candidates knew how to find the gradient of a perpendicular line, but those using $y = mx + c$ made more arithmetic slips than those using the more appropriate form

$$y - y_1 = m(x - x_1).$$

Part (b) Apart from those who used the wrong pair of equations, this part was usually answered correctly.

Part (c) Although this part was meant to be challenging, there were many successful attempts, particularly by those who used a sketch and reasoned on a 3, 4, 5 triangle. It had been intended that candidates would have formed an equation such as $16 + (k + 2)^2 = 25$, but more commonly something such as $16 + y^2 = 25$ was seen, resulting in the incorrect values of ± 3 .

Q13.

Part (a)(i) Some candidates ignored the request to state the coordinates of B even though they were using the height of the triangle as 5. The negative x -coordinate of A caused

quite a few to feel that the triangle had a negative area. Far too many when finding

$\frac{1}{2} \times 1 \times 5$ wrote the answer as 3.

Part (a)(ii) Practically every candidate found the correct integral although some made errors when cancelling fractions.

Part (a)(iii) It was necessary here to have the lower limit as -1 and the upper limit as 0 . Many reversed the order and by some trickery arrived at a positive value. This was penalised and so very few, even though many had an answer of 1 for the area, scored full marks for this part of the question.

Part (b) Most candidates differentiated correctly but, because of poor understanding of negative signs, many wrong values of -13 were seen for the gradient. There is obviously confusion for many between tangents and normals and several thought the gradient of the

tangent was $\frac{-1}{17}$.

Q14.

This calculus question again proved to be a good source of marks for many candidates.

Part (a)(i) was answered very well but in part (a)(ii) a significant minority used $\frac{d^2x}{dx^2} = 0$

instead of $\frac{dy}{dx} = 0$ dx^2 . In part (a)(iii), although most candidates showed that they understood the method to find the gradient of the tangent, a significant proportion failed to find and use the gradient of the normal. Many candidates showed that they had a good understanding of integration but, surprisingly, errors in integrating $(x + 1)$ were almost as

common as errors in integrating $\frac{4}{x^2}$. Examiners expected to see some simplification of $\frac{4}{x^2} + 4x^{-1}$ the algebraic term $\frac{-1}{x^2}$. In the final part, the evaluation of $F(4) - F(1)$ as 8.5 was the most common error.

Q15.

In part (a)(i), many candidates were unable to produce a correctly expressed solution, despite dealing correctly with the differentiation. The main error was the omission of brackets around

$8x + 3$. Part (a)(ii) was well answered by the majority of candidates. Part (b)(i) was well

answered by the majority of candidates, although a common error was $6y^2 + \frac{1}{x}$. Most candidates correctly substituted $y = 1$ and obtained 7 , although 6 was a common evaluation; a substitution of $y = 2$ was also seen. Many candidates who obtained 7 then proceeded, incorrectly, to write $(y - 1) = 7(x - 2)$ although a few candidates correctly wrote

$\frac{dx}{dy}$ “ $(x - 2) = 7(y - 1)$ ”. The significance of $\frac{dx}{dy}$ was rarely appreciated, with a significant

number of candidates finding an incorrect gradient of $-\frac{1}{7}$, and confusing $\frac{dx}{dy}$ with the gradient of a normal.

Q16.

Part (a)(i) was well answered, with the main error being $\int -e^x dx = -2e^{2x}$. In part (a)(ii) the answer was given. Frequently candidates tried to 'fudge' their solutions to arrive at the correct result, with common error being $4\ln 2 - 2 - \frac{1}{2} = 4\ln 2 - \frac{3}{2}$, the given answer.

Part (b)(i) was again well answered by the majority of candidate, although in part (b)(ii) the step from $2x = \ln 4$ to obtain $x = \ln 2$ was often unconvincing. Some candidates answered part (c) very well and gained full credit. The main error was using 8 or -8 as the gradient of the normal, however many candidates earned at least partial marks.

Although correct answers were seen in part (d), they were not common. Many candidates earned the method mark by the substitution of $x = 0$ and some went on to find the area of the triangle. The final mark was often lost through a failure to realise that the area was positive.

Q17.

This was the most successful question on the paper with most candidates gaining full marks or close to it. Simple sign and coefficient errors were made in the differentiation, but the vast majority of candidates demonstrated they could use the chain rule correctly. Only a few tried to find y in terms of x first, doing the work actually required for part (c). Some candidates found an equation of the tangent instead of the normal, and a few left t in their final answer. A form of the answer was not specified so any correct form was accepted, but some working with the $y = mx + c$ form sometimes found a value for c , but did not write down an equation as their final answer.

Similarly in part (c) a correct, unsimplified form was accepted for full credit, although it was noted that many did go on to make algebraic errors in attempting to simplify their equation. Most candidates did know what was required for a cartesian equation and many alternative correct versions were seen or a simple sign error was made. A few candidates thought they were required to find the tangent, and some seemed to invent their own equation, such as an ellipse, to substitute into.

Q18.

Part (a) of this question was well answered, most candidates being familiar with the equation $y = ax^b$ and the technique needed to convert it into linear form. Some candidates seemed less happy with part (b), but most managed to show enough knowledge to score well here. Errors often arose from confusion between the intercept 1 on the vertical axis and the corresponding value of y , which required the taking of an antilogarithm.

Q19.

Part (a)(i) Most candidates were able to show that the gradient was $-\frac{3}{2}$. However, examiners had to be vigilant since fractions such as $\frac{6}{4}$ and $\frac{-4}{6}$ were sometimes equated to $-\frac{3}{2}$.

Part (a)(ii) Many candidates did not heed the request for integer coefficients and left their answer as $y = -\frac{3}{2}x + 8$. Many who attempted to express the equation in the required form were unable to double the 8 and wrote their final equation as $3x + 2y = 8$.

Part (b)(i) Most candidates realised that the product of the gradients should be -1 . However, not all were able to calculate the negative reciprocal. Others used the incorrect point and therefore found an equation of the wrong line.

Part (b)(ii) Many candidates made no attempt at this part of the question. The most successful method was to substitute $y = 7$ into the answer to part (b)(i) or to equate the gradient to $\frac{2}{3}$.

There were also some good answers using a diagrammatic approach. Those using Pythagoras usually made algebraic errors and so rarely reached a solution.

Q20.

Part (a) Most candidates found the correct coordinates of the centre, although some wrote these as $(3, -2)$ instead of $(-3, 2)$. Those who multiplied out the brackets were often unsuccessful in writing down the correct radius of the circle.

Part (b)(i) Most candidates were able to verify that the point N was on the circle, although some, who had perhaps worked a previous examination question, were keen to show that the distance from C to N was less than the radius and that N lay inside the circle.

Part (b)(ii) Most sketches were correct, though some were very untidy with several attempts at the circle so that the diagram resembled a chaotic orbit of a planet. Some candidates omitted the axes and scored no marks.

Part (b)(iii) The majority of candidates found the gradient of CN and then assumed they had to find the negative reciprocal of this since the question asked for the normal at N . Reference to their diagram might have avoided this incorrect assumption.

Part (c)(i) Most wrote $PC^2 = 5^2 + 4^2$, provided they had the correct coordinates of C . However, the length of PC was often calculated incorrectly with answers such as $\sqrt{31}$ and $\sqrt{36} = 6$ seen quite often.

Part (c)(ii) Although there were many correct solutions seen, Pythagoras' Theorem was often used incorrectly. A large number of candidates wrote the answer as a difference of

two lengths such as $\sqrt{41}-5$. Candidates need to realise that obtaining the correct answer from incorrect working is not rewarded; quite a few wrote $\sqrt{41}-\sqrt{25}=\sqrt{16}=4$ and scored no marks. Many who drew a good diagram realised that a tangent from (2, 6) touched the circle at (2, 2) and so the vertical line segment was of length 4 units.

Q21.

This question proved to be a good source of marks for many candidates although the

expansion in part (b) was quite frequently incorrect with $\left(\frac{2}{x}\right)^2$ being written as $\frac{4}{x}$ or as $\frac{2}{x^2}$. Most candidates displayed good differentiation skills in part (c) and even more realised that the answer to part (d) required evaluation of their answer to part (c) when $x = 2$. Even though the majority of candidates knew how to find the equation of the normal it was very common to find a mark lost in the final part of the question because candidates were unable to rearrange their answer in to the form stated in the question.

Q22.

(a) (i) Candidates used various methods to prove that $k = -2$. Some used the most direct method of substituting $x = 5$ into the given line equation and solving for y ; some chose to verify that $x = 5$ and $y = -2$ satisfied the equation of the straight line. Others took a longer route; they found the gradient using (1, 1) and (5, -2) and then found the equation passing through one of the points and proved it to be the given one.

(ii) Most candidates knew how to find the midpoint of a line. A few made a simplification error and wrote $\left(3, \frac{1}{2}\right)$ instead of $\left(3, -\frac{1}{2}\right)$. The common error amongst the weaker candidates was to subtract the coordinates instead of adding them.

(b) Many candidates gave fully correct answers here. However, some, having obtained $\frac{-2-1}{5-1}$ wrote $\frac{3}{4}$ as a final answer. A few candidates used $\frac{x_1-x_2}{y_1-y_2}$.

(c) (i) Most knew the gradient rule for perpendicular lines. However, not all could implement it since it involved the reciprocal of a fraction.

(ii) At least half of the candidates found the equation of the line passing through the midpoint of AB instead of through C . (iii) Most realised the need to substitute $y = 0$ into their AC equation and solve for x , so they at least earned the method mark. Even those with the correct equation did not always earn two marks. Some had difficulty in simplifying $1/3 \div 4/3$.

Q23.

This final question was a good source of marks for many candidates. Even the weaker

candidates were able to give correct answers for parts (a) and (c) which tested standard C2 work on differentiation and integration. A significant minority of candidates started part (b)(i) by equating dy/dx to zero instead of substituting $x = 0$ to find the value of the derivative. Those who were able to answer the first two parts of part (b) correctly generally went on to find the correct coordinates of P . A significant number of candidates left their final answer as 24.3 without considering the area of the triangle OAP and carrying out the subtraction. For those candidates who attempted to find the area of the triangle some applied very lengthy methods which involved finding the length of each side of the triangle

then using the cosine rule to find angle OPA then using the formula $\frac{1}{2} ab \sin C$ or involved the integration of the equations of the two

lines. To their credit a number of such candidates did avoid premature approximation to reach the correct value for the area of the triangle which could have been obtained more

directly by using $\frac{1}{2} OA \times |y_P|$.

Q24.

The great majority of candidates knew what was expected in part (a) but some made

errors in attempting $\frac{d}{dt}\left(\frac{2}{t}\right)$; sign and coefficient errors being common and even $\ln t$ was seen.

The chain rule was usually used correctly although a few multiplied the two derivatives or had the expression 'upside down'. Errors were sometimes made in simplifying the expression, with a sign error or 2 or 8 being seen instead of $1/2$, or the t^2 term moving to the top line.

Candidates generally showed they knew how to find the equation of a straight line in part (b) using their gradient from part (a) although a few found the normal rather than the tangent. Some candidates lost a mark through not giving their final answer in the requested form, or did not notice that their a , b , c were not integers, particularly those few who had a $\ln$ term in part (a). Part (c) was usually started well but not always finished convincingly, as the answer was given and sufficient working not shown. The question did request a verification of the given answer rather than requiring it to be shown, but few

candidates did it by the most efficient method of demonstrating $-4t \times \frac{2}{t} = -8$. There were many good answers using the elimination of t , although some candidates were too quick to write down the given answer without showing where it had come from.

Q25.

Part (a)(i) Although most obtained the correct gradient, some omitted the negative sign (particularly those who relied on a sketch for their evaluation) and some had a fraction with the change in x as the numerator which immediately scored no marks. Quite a few found mid-points (possibly since that had appeared on previous examinations) and others added the coordinates instead of finding the differences in their quotient expression for the gradient.

Part (a)(ii) The use of their gradient to obtain the given equation was the most successful method. Those using $y = mx + c$ had a tendency to introduce a new ' c ' by doubling both

sides but then substituted their value back into the original equation. The most successful candidates used the formula $y - y_x = m(x - x_x)$. Some re-arranged the given equation to check the gradient then checked one set of co-ordinates; others checked two points and indicated that a straight line has the form $ax + by = c$.

Part (b) Those using substitution often began by using an incorrect rearrangement of one of the equations. If they attempted elimination, sometimes only part of an equation was multiplied by the appropriate constant. Many added the equations instead of subtracting. Of those who wrote $14y = -7$, just as many obtained an incorrect answer of $y = -2$ as the correct answer of $y = -1/2$.

Part (c) The condition for perpendicularity was generally known but some were unable to evaluate -1 divided by -1.5 . A few omitted the $-$ sign while some referred to the equation $3x + 2y = 17$ and gave a gradient of $1/3$. Once again, those determined to use $y = mx + c$ often made errors in the constant due to the fractional coefficient of x . Quite a few did not use the point s as instructed, choosing to use the point C instead.

Q26.

Part (a) Most candidates answered this part correctly. A few included the 7 or thought the derivative of the first term was $7x$ and the $-$ sign was sometimes lost.

Part (b) Many substituted $x = 1$ correctly, though it was apparent that they did not recognise this value of -10 to be the gradient of the tangent. Many correctly found y as 5 but stopped there. Again, some correct attempts at the tangent equation using $y = mx + c$ foundered and quite a large number attempted to find the equation of the normal.

Part (c) Use of the value of $\frac{dy}{dx}$ was the only acceptable method here. Evaluations of y at different points or finding the second derivative were common but earned no marks.

Q27.

- (a)
 - (i) It was not uncommon to see the gradient written as $-3/-12$, which was then cancelled down to $-1/4$ or 4 . However, most candidates were successful in finding the correct gradient.
 - (ii) Most candidates found a correct equation for the line AB but were unable to rearrange it into the given form with integer coefficients. Far too many mistakes are made when candidates rely on using an equation of the form $y = mx + c$ and they may be better advised to remember alternative forms for the equation of a line.
- (b) It had been expected that solving a pair of simultaneous equations would have been routine at this level. It was, therefore, quite disappointing to see the number of arithmetic errors preventing many from finding $(7, 1)$ as the point of intersection.

Q28.

- (a) Most candidates knew how to differentiate, but a few made careless mistakes when

differentiating $3x$ and others insisted on adding $+ C$ to their otherwise correct answer.

- (b) (i) Usually sufficient working was shown to verify that the gradient at P was equal to 5.
- (ii) Many seemed confused between the tangent and the normal, with many clearly trying to find the equation of the former rather than the latter. The major problem, once again, was rearranging the equation into one involving integer coefficients.
- (c) It was clear that some candidates were unfamiliar with the need to consider the sign of dy/dx to determine whether y was increasing or decreasing. It is a pity that $\frac{d^2y}{dx^2}$ also had value -16 when $x = 1$, yet it was surprising to see how many felt they had to consider the sign of the second derivative rather than the first.

Q29.

Parts (a)(i) and (a)(ii) were generally well answered although it was surprising to find the derivative of x seemingly causing as many problems as the derivative of $2/x$. Although the method for finding the equation of a line was generally well known, a significant minority of candidates found the equation of the tangent rather than the equation of the normal.

Q30.

This was a straightforward question on coordinate geometry intended to get the candidates off to a confident start and many scored full marks. However many made very fundamental errors.

Part (a) Instead of finding the arithmetic mean of the pair of x -coordinates and the pair of y -coordinates, many candidates found the difference of the coordinates before dividing by 2. A few were able to deduce the coordinates of the mid-point from a diagram and some used the half line vector approach successfully.

Part (b) There were many correct answers but a small number were incorrect and some candidates wrote $\sqrt{52} = 4\sqrt{13}$. Notational errors such as $36 + 16 = \sqrt{52}$ were seen and will not always be condoned. It was surprising how many candidates made no attempt at the distance formula.

Part (c)(i) Although the majority of answers were correct, an alarming number of candidates could not simplify $-6/-4$ correctly. It was also very disappointing to see many using difference in x values/difference in y values in order to find the gradient of a line.

Part (c)(ii) Candidates were expected to use their gradient to determine the equation of the line and many had a complete proof. It was not expected that so many would use the equation $y = mx + c$ at this stage. Many fudged their work to get the printed answer. Some imagined that it was sufficient simply to show the required line had the same gradient as they had evaluated in part (c)(i) or to show that **one** of the points satisfied the given equation. Neither of these on their own scored any marks. Verifying two of the facts was given some credit.

Part (d) Although the majority of candidates found the point of intersection, there were many algebraic and sign errors in the attempts seen – adding equations after equalising coefficients being the most common. Many students did not recognise what was required and made no attempt at it.

Q31.

Part (a) The equation of a circle seemed widely known, although some candidates wrote things such as $C = 2\pi r$. Sign errors were common and some forgot to square the radius or wrote $\sqrt{5}$ on the right side of the equation.

Part (b) Most candidates realised that the easiest way to verify that the point P lies on the circle was to substitute $x = 6$ and $y = 2$ into the left hand side of the equation and to show this was equal to 25, the same as the right hand side. Other ingenious methods were used in addition to verifying that PC was equal to 5.

Part (c) The same errors relating to part (c)(i) of question 1 were repeated here. Most candidates, however, were successful in finding the correct gradient of CP.

Part (d)(i) The relationship that the product of the gradients is -1 seems well known but a number of arithmetic slips occurred when finding the gradient of the perpendicular line.

Part (d)(ii) Several ignored the hint of part (d)(i) and many failed to realise that the tangent was a line passing through P(6, 2). It was not necessary to simplify a correct form of the equation in order to score full marks. Those who used $y = mx + c$ usually fared badly here, because they rarely obtained a correct form of the equation.

Q32.

This question proved to be a good source of marks for many candidates. In part (a)(i) most candidates quoted the correct value for the power p. Although many candidates produced correct answers for the integration, all the usual errors, as illustrated by

$$\frac{3}{2}x^{1.5}, x^{1.5}, \frac{x^{-0.5}}{-0.5}, \text{ and } \frac{\sqrt{x}^2}{2},$$

EXAM PAPERS PRACTICE

were seen. Most candidates stated and used the correct limits in part (a)(iii) but evaluation was sometimes poor. Many of the candidates found a correct equation for the tangent but it was surprising to find a significant minority of these candidates either making no attempt to find the area of triangle AOB or giving the area as a negative value. For full marks in part (c) the examiners were expecting to see the word 'translation'. Some candidates wrote 'tr' but this was not considered to be clear enough to distinguish between 'transformation' and 'translation'. Common wrong answers involved 'stretch' and vectors in the wrong direction. In part (d) there was a significant minority of

candidates starting their solution with the incorrect statement $\sqrt{x-1} = \sqrt{x} - \sqrt{1} = \sqrt{x} - 1$ but in general the trapezium rule was well understood and many candidates were able to give the final answer to the correct required degree of accuracy in this numerical integration question.